

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

CE-7102-GS
B.E. VII Semester
Examination, December 2020
Grading System (GS)
Traffic Engineering
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. What are the different vehicular characteristics which affect the road design?
2.
 - a) Discuss in detail about objectives and scope of traffic engineering.
 - b) Describe the need for O-D survey and discuss the survey methods.
3.
 - a) Explain PIEV theory in detail.
 - b) How does a vehicles speed and braking capacity affects the road user characteristics?
4.
 - a) Write a note on:
 - i) Practical capacity of road
 - ii) Level of service
 - b) How Traffic volume survey is carried out? Explain presentation of data obtained.

CE-7102-GS

PTO

[2]

5. a) An intersection controlled by traffic signals is formed by two roads running North to South and East to West. There is a heavy turning movement of 500 vehicles per hour from South to East. The flow from North to South is 800 vehicles per hour through an approach width at stop line of 8m. The cycle time is 70 seconds and the effective green time for the flow from the North is 32 seconds. The effective right turning saturation flow is 450 vehicles per hour for an opposing flow of 800 vehicles per hour, related to a minimum head-way of 2.5 seconds for the right turning stream. Calculate the early cut off period required to allow for the right-turning movement.
b) Explain merits and demerits of one way street.
6. Discuss “Road Accident Audit”.
7. a) As part of a class project, a group of students collected a total of 120 spot speed samples at a location and determined from this data that the standard deviation of speeds was 6 kmph. If the project required that the confidence level be 95% and the limit of acceptable error was 1.5 kmph, determine whether these students satisfied the project requirement.
b) Describe the method of collection of parking data and analysis of parking data.
8. a) Discuss the problems and remedial measures of mass transportation.
b) What are free ways? Discuss the advantages of free ways over express ways.
